Building Cooperative Robot Behavior from Basic Action Modules

An LLM-driven control framework for two cooperating AR4 arms

Jan M. Straub

Supervisor: Prof. Dr. Artur Andrzejak

July 22, 2026

Institute for Computer Science, Artificial Intelligence for Programming
Ruprecht-Karls-University of Heidelberg

Overview

1 **The problem**
The reasoning–execution gap, and why cooperation is the hard case

2 **Pillar 1: Grounding**
A 20-operation vocabulary makes a small, local VLM reliable

3 **Pillar 2: Cooperation**
Handoffs and parallelism composed by the model

4 **Pillar 3: The Boundary**
The one failing task, what the ablations really show, limitations

5 **Conclusion**
The five contributions, the path to hardware, future work

Condensed Results

Thesis in one sentence

A **locally hosted vision-language model (VLM)**, grounded in a structured pool of **20 basic operations**, turns natural language into reliable **robot behavior**, with **no fine-tuning**.

Solved

Success on all single-robot tasks and dual-arm handoffs with the default backend.

Composed

Cooperation is composed at runtime from general primitives, not a hard-coded. Negotiation adds explicit reasoning, but lifts only the default model.

Gap

Concurrent bimanual lift fails: target selection is not partner-aware, plus a simulator attachment limit.

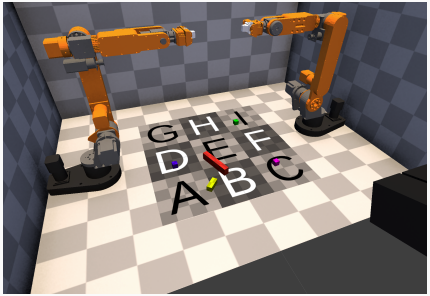
The Problem

The Reasoning–Execution Gap

- Classical robots: precise, but caged. Rigid paths, no tolerance for changes.
- VLMs bring world knowledge and reasoning, but cannot reliably turn abstract reasoning into *physical* action.
- Models often ignore past physical state when planning the next move.

Why cooperation is the hard case

Two arms must **avoid collision**, **share a workspace**, and **synchronize timing**.



Two AR4 [1] arms, a shared workspace, and a stereo camera in Unity [2].

Two Design Commitments

Run the model *locally*

- No cloud cost, no external dependency.
- No sensitive scene data leaving the machine.
- A compact VLM is enough; no frontier model required.

Ground the model in *modular operations*

- Reasoning restricted to a vocabulary of *executable* actions.
- Separates semantic understanding from physical execution, making it reliable.
- Vision resolves targets live; no task-specific training.

Default backend model: **Magistral Small 2509 (24B, 4-bit) [3]**

Background

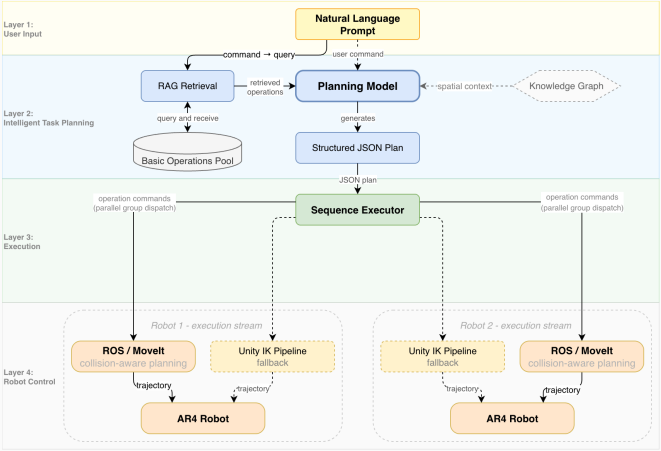
- **End-to-end VLA** (RT-2, PaLM-E) [4, 5]: great generalization, but data-hungry and opaque.
- **Code-as-Policies** [6]: compositional and transparent, but tied to its API set.
- **Framework-scale** (AutoRT) [7]: practical scaffolding, but no joint reasoning over a *shared* object.
- **RoCo** [8]: per-robot agents negotiate, but free-form way-points.

Research gap

Runtime composition of shared-object coordination from a fixed vocabulary of *operations*, cooperation *grounded* in reasoning rather than a fixed execution path, is comparatively unexplored.

Pillar 1: Grounding

System Architecture



Built on Unity, ROS 2 [9], and MoveIt [10].

The Operation Pool

Operations organized by **complexity**, four tiers:

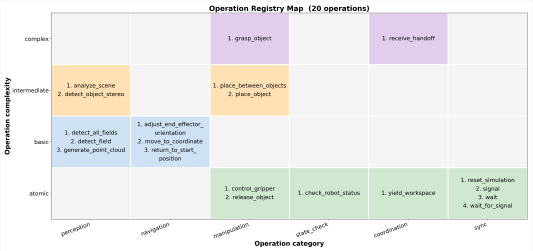
Atomic (8) indivisible, no perception: gripper, signal, wait.

Basic (6) single dispatch, introduces perception.

Intermediate (4) composite single-robot: placement, stereo detection, scene analysis.

Complex (2) full pipelines: grasp planning, handoff reception.

Why it grounds: model can only express plans the robot can actually run.



Reflection Recovery

Gated retry on failure

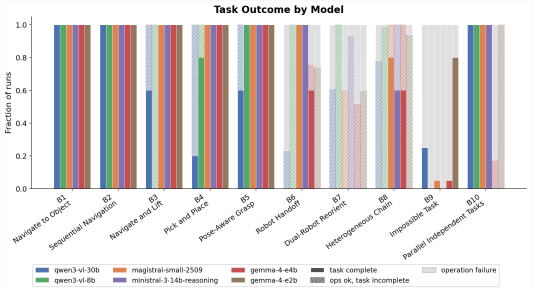
- On an eligible failure, re-query the model with a structured *hint* (Reflexion-style verbal feedback [11]), closing the perception–action loop at the language level.
- Eligibility is gated to Navigation / Manipulation / Perception: no point re-querying a sync barrier.

An optional knowledge-graph layer (spatial reachability / handoff queries) is also available; its ablation cost is covered later.

Grounding ⇒ Reliability

- **B1–B5** (navigate → grasp → full pick-and-place): **success over all 25 runs**, zero hallucinated ops, zero retries.
- *Identical* operation sequence on all 5 runs of each benchmark.
- **B8** (long chain): 20 sub-tasks, **85 operations**/run, 5 cycles, all executed, no cascading errors.

Cost is dominated by motion primitives (`grasp_object`, `place_object`).



Magistral, Ministral, and both Gemma 4 models clear the full single-robot gradient; the Qwen3-VL models fail from B3 onward by truncating plans.

Pillar 2: Cooperation

The Model Composes

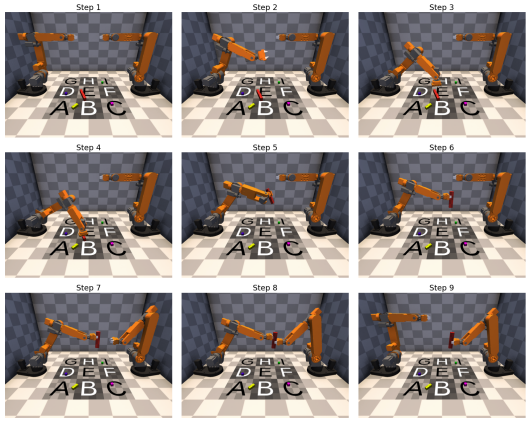
Two general primitives

- `signal` / `wait_for_signal`: one robot pauses at a checkpoint until the other signals readiness.
- **Parallel groups**: operations dispatched concurrently when no ordering is needed.

The executor holds **no** coordination policy. Guided by prompt rules and retrieved examples, the model places the synchronization barriers and parallel groups; the physical coordination, approach geometry and release timing, lives in the operations, not the plan.

The Handoff

- Magistral and Ministral: 5/5 runs; all operations first try.
- Ordering and the signal/wait barrier are set by an prompt rule.
- Gemma-4B partial (3/5); Qwen3 and Gemma-2B fail at plan construction.
- **Parallelism**: the same primitives schedule independent pick-and-places (4/6 backends).



Handoff sequence.

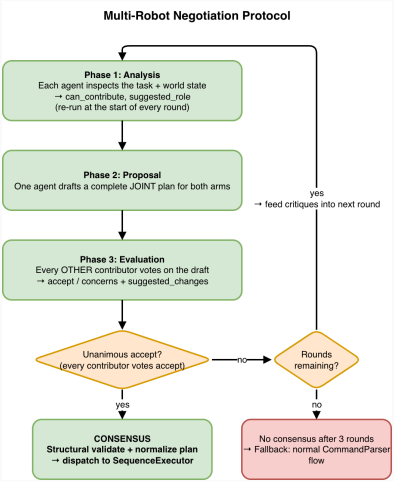
Negotiation

A single model planning both arms under-specifies roles; *implicit conflicts* surface only at execution.

Protocol:

1. **Analyze**: each agent assesses its own role (parallel).
2. **Propose**: one agent/round drafts a *complete* plan (round-robin).
3. **Evaluate**: peers accept/reject with justification; **unanimity** required. Rejections feed the next proposal.

No consensus in 3 rounds → single-model parsing.



Negotiation Pays Off

	Task SR	Ops/run
Negotiation on	**0.713** (57/80)	35.8
Negotiation off	0.613 (49/80)	29.9

- **+10 pp** task success for the **default model**.
- Roughly neutral for three backends; *negative* for two (-19, -21 pp).
- Pays off when the single-shot plan is already close to correct; doesn't rescue weaker planners from role conflicts.

Pillar 3: The Boundary

Concurrent Bimanual Lift

All six models fail. Three failure modes:

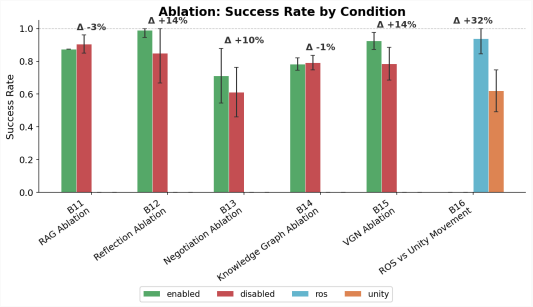
1. Signal primitives never dispatched once control crosses robots outside a parallel group.
2. Final lift: two *independently* chosen targets land too close together, violating the separation check.
3. Deeper sim limit: a Unity `Transform` has one parent, so a single object can't attach to two grippers at once.

Gap

Partner-aware target selection. Each arm picks a valid target; nothing checks one *against the other*.

Ablation Synthesis

- **ROS/MoveIt +32 pp**: Unity-only grasp timeouts; plus collision-aware planning.
- **Reflection +14 pp**: clears boundary-reach failures; doubles time when it fires.
- **Negotiation +10 pp**: default model only; neutral/negative on the other five.
- **VGN +14 pp** (+35 on grasp-held): center-of-object grasp, not predicted orientation.
- **RAG / KG: net cost**: prompt over-anchoring, helps no backend.



Per-component deltas (default model; $n = 20$ for B11–B13, B16; $n = 5$ for B14–B15).

Limitations

Sensing

Cube features too small to match, so the stereo cloud is sparse.

Latency

VLM parse $\approx$ **20 s** median, which prevents *reactive* control; re-plan only between motions, never during.

Sim-to-real

Robot joint values all tuned for simulation, so they need revalidation on hardware.

Validation

5/6 models forward impossible inputs; the guard must be pre-parse.

Conclusion

Five Contributions

1. **Grounded control framework**: a hierarchical 20-operation registry, retrieved on demand, turning language into verified behavior with no fine-tuning.
2. **Runtime-composed coordination**: the model *sequences* handoff steps and parallelizes independent tasks with `signal`/`wait` and parallel groups; the physical transfer itself is an engineered operation it invokes.
3. **Consensus-gated negotiation**: helps only an already-capable planner (+10 pp default; neutral-to-negative for the rest).
4. **Operation-gated reflection**: gated verbal repair; +14 pp for the default model.
5. **AutoRT safety adaptation**: a hard kinematic gate the semantic check can't override.

Future Work

Deployment

- `-env real` swaps camera and hardware adapters at the abstraction boundary.
- Control switches to ROS/MoveIt collision-aware planning, already dual-robot configured.

Remaining work

- **Depth**: LIDAR or per-robot stereo unlocks VGN-grade grasping; a SLAM volumetric map adds object permanence.
- **Force-torque**: a wrist sensor replaces the contact-duration heuristic.
- **Bimanual**: partner-aware target selection closes the boundary gap.
- **Pre-parse validation**: robot-ID / object-existence checks before any model call.
- **Faster inference**: distilled task-specific models or speculative decoding to shrink the 20 s window.

Demo & Questions

What works

- Single-robot: works in 25/25 runs.
- Long chain: 85 ops/run, 5 cycles, no cascading errors.
- Handoff: works and correctly realized sync barrier.
- Parallelism: works for 4/6 backends.
- Safety gate: no false-accept, reproducible.

What each component contributes

- VGN grasp: +35 pp Magistral, +33 pp pooled.
- ROS/MoveIt: +32 pp, Unity-only grasp timeouts.
- Reflection: +14 pp task.
- Negotiation: +10 pp, default model only.
- RAG: model-dependent; KG (B14): small cost.

The one open gap

- Concurrent bimanual B7: all six models fail.
- Cause: partner-aware target selection.

Thank you.

`github.com/JanMStraub/Cooperative-Robot-Behavior-from-Basic-Action-Modules`

References

[1] C. Annin, "Annin robotics," Accessed: 2026-06-09, 2026. [Online]. Available: https://anninrobotics.com

[2] Unity Technologies, "Unity," 2026, game development platform. [Online]. Available: https://unity.com/

[3] Mistral AI, "Magistral-small-2509," https://huggingface.co/mistralai/Magistral-Small-2509, 2025, large Language Model.

[4] A. Brohan, N. Brown, J. Carbajal *et al.*, "Rt-2: Vision-language-action models transfer web knowledge to robotic control," 2023. [Online]. Available: https://arxiv.org/abs/2307.15818

[5] D. Driess, F. Xia, M. S. M. Sajjadi *et al.*, "Palm-e: An embodied multimodal language model," 2023. [Online]. Available: https://arxiv.org/abs/2303.03378

[6] J. Liang, W. Huang, F. Xia *et al.*, "Code as policies: Language model programs for embodied control," 2023. [Online]. Available: https://arxiv.org/abs/2209.07753

[7] M. Ahn, D. Dwibedi, C. Finn *et al.*, "Autort: Embodied foundation models for large scale orchestration of robotic agents," 2024. [Online]. Available: https://arxiv.org/abs/2401.12963

[8] Z. Mandi, S. Jain, and S. Song, "Roco: Dialectic multi-robot collaboration with large language models," 2023. [Online]. Available: https://arxiv.org/abs/2307.04738

[9] S. Macenski, T. Foote, B. Gerkey *et al.*, "Robot operating system 2: Design, architecture, and uses in the wild," *Science Robotics*, vol. 7, no. 66, May 2022. [Online]. Available: http://dx.doi.org/10.1126/scirobotics.abm6074

[10] D. Coleman, I. Sucan, S. Chitta *et al.*, "Reducing the barrier to entry of complex robotic software: a moveit! case study," 2014. [Online]. Available: https://arxiv.org/abs/1404.3785

[11] N. Shinn, F. Cassano, E. Berman *et al.*, "Reflexion: Language agents with verbal reinforcement learning," 2023. [Online]. Available: https://arxiv.org/abs/2303.11366

[12] M. Breyer, J. J. Chung, L. Ott *et al.*, "Volumetric grasping network: Real-time 6 dof grasp detection in clutter," 2021. [Online]. Available: https://arxiv.org/abs/2101.01132

Additional Slides

Backup: Bimanual Lift per-run Outcomes

Run	Ops ok	Dur (s)	Failing op
1	4/8	111	`wait_for_signal`
2	4/10	43	`wait_for_signal`
3	6/7	211	`move_to_coordinate` (collision)
4	6/7	206	`move_to_coordinate` (collision)
5	4/10	46	`wait_for_signal`

Two clusters: mis-realized barrier, never executed (1,2,5), and lift-target collision at $\sim$0.10 m (3,4).

Backup: Prompt-Driven Parallelism

General rule in the prompt

```
Same parallel_group = concurrent.
Later group waits for all prior.
VAR DEPENDENCY LAW: if B reads a
$var captured by A, B must be in a
strictly higher parallel_group.
```

B10 plan the model emitted

```
g1:  detect(R1,blue) || detect(R2,green)
g2:  grasp(R1) || grasp(R2)
g3:  field(R1,A) || field(R2,B)
g4:  place(R1) || place(R2)
```

- 4/6 models: 40/40 operations, zero retries.
- Each grasp waits only on its own detection ($var dependency); the two arms otherwise run fully in parallel.

Takeaway

General rule in → task-specific parallel schedule out, with no matching example. The handoff, by contrast, follows an explicit ordering rule in the prompt.

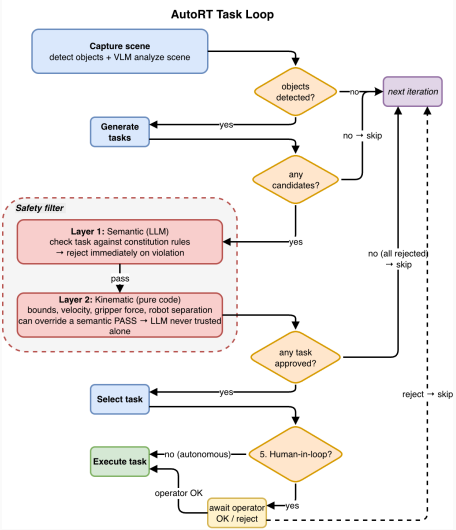
Backup: AutoRT Adaptation & Safety Gate

Re-targeted AutoRT for *fixed-base* manipulation.

Two-layer safety (Robot Constitution):

- Semantic VLM check
- Hard kinematic check the semantic layer can't override.

Result: never admits an unsafe task. Slightly over-rejects aggressive-but-safe phrasing.



Backup: What VGN [12] changes

Held & lifted	VGN on	VGN off
Magistral	**14/20 (0.700)**	7/20 (0.350)
Pooled, 6 models	**0.750**	0.417

- Every backend improves: +20 pp (Ministral) to +65 pp (Gemma-4B).
- Success = held and lifted clear of the table.

Mechanism: position, not orientation

- **Same orientation both conditions**: top-down, yaw from the object's long axis. VGN's own poses fail motion planning and are discarded.
- **The difference is position**: VGN grasps the detected object *center*; the baseline's contact point sits off-center on angled cubes and slips.

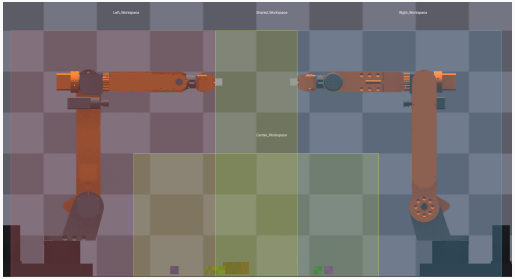
Scope

Position comes from the WorldState oracle, so B15 isolates grasp *positioning*. Recovering VGN's learned pose needs a real depth sensor.

Backup: Collision Safety Layers

1. **Static pre-check** (Python, before dispatch): `CoordinationVerifier` checks workspace bounds, object conflicts, deadlock detection on the planned targets.
2. **Workspace regions**: left / right / shared zones; a contested zone is handed over explicitly via `yield_workspace`.
3. **Runtime guard** (Unity, every physics step): `ProximityGuard` freezes motion at 0.25 m end-effector distance, resumes at 0.35 m; link-to-link limit 0.15 m.

The LLM is never trusted for safety: layers 1 and 3 are code, not reasoning.



Backup: Single-Robot Operation Latencies

Operation	Bench	n	Mean (s)
`detect_object_stereo`	B1–B5	30	0.80
`move_to_coordinate`	B1–B3	20	4.37
`grasp_object` (B3)	B3	5	19.84
`grasp_object` (B4)	B4	5	10.93
`place_object`	B4	5	34.43

Perception is cheap and stable; contact-rich manipulation dominates and varies with object pose.

Backup: Model × Task Matrix